HDOC DAY-13

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Batch - 15

1. Algorithm

1. Start.
2. Declare variables for the number, temporary number, digit, choice, counters, and sums.
3. Input a positive integer `num`.
4. If `num <= 0`, display an invalid-input message and terminate.
5. Display the menu:
   - `1` – Count even and odd digits.
   - `2` – Find sum of even and odd digits.
6. Input the user's choice.
7. Store `num` in `temp`.
8. Use a `switch` statement:
   - **Case 1:**
     1. Repeat while `temp > 0`.
     2. Extract the last digit using `digit = temp % 10`.
     3. If `digit % 2 == 0`, increment `even_count`; otherwise increment `odd_count`.
     4. Remove the last digit using `temp = temp / 10`.
     5. Display `even_count` and `odd_count`.
   - **Case 2:**
     1. Repeat while `temp > 0`.
     2. Extract the last digit.
     3. If the digit is even, add it to `even_sum`; otherwise add it to `odd_sum`.
     4. Remove the last digit.
     5. Display `even_sum` and `odd_sum`.
   - **Default:** Display `"Invalid choice!"`.
9. Stop.

2. Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Number = 12045, Choice = 1	Even_dig = 3, Odd_dig = 2	Even_dig = 3, Odd_dig = 2	**Pass**
2	Number = 12345, Choice = 2	Sum_even = 6, Sum_odd = 9	Sum_even = 6, Sum_odd = 9	**Pass**
3	Number = 24680, Choice = 1	Even_dig = 5, Odd_dig = 0	Even_dig = 5, Odd_dig = 0	**Pass**
4	Number = 13579, Choice = 2	Sum_even = 0, Sum_odd = 25	Sum_even = 0, Sum_odd = 25	**Pass**

3. C Program

```
#include <stdio.h>

int main()
{
    long long num, temp;
    int choice, digit;
    int even_count = 0, odd_count = 0;
    int even_sum = 0, odd_sum = 0;

    printf("Enter a positive integer: ");
    scanf("%lld", &num);

    if (num <= 0)
    {
        printf("Invalid input! Please enter a positive 
integer.\n");
        return 0;
    }

    printf("\nMENU\n");
    printf("1. Count even and odd digits\n");
    printf("2. Find sum of even and odd digits\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    temp = num;

    switch (choice)
```

```
    {
        case 1:
            while (temp > 0)
            {
                digit = temp % 10;

                if (digit % 2 == 0)
                    even_count++;
                else
                    odd_count++;

                temp = temp / 10;
            }

            printf("Even_dig = %d\n", even_count);
            printf("Odd_dig = %d\n", odd_count);
            break;

        case 2:
            while (temp > 0)
            {
                digit = temp % 10;

                if (digit % 2 == 0)
                    even_sum = even_sum + digit;
                else
                    odd_sum = odd_sum + digit;

                temp = temp / 10;
            }

            printf("Sum_even = %d\n", even_sum);
            printf("Sum_odd = %d\n", odd_sum);
            break;

        default:
            printf("Invalid choice!\n");
    }

    return 0;
}
```

4. Compilation and Execution

```
aaryanrajput@aaryans-macbook ~ % nano day13.c
aaryanrajput@aaryans-macbook ~ % clang day13.c -o day13.c
aaryanrajput@aaryans-macbook ~ % ./day13.c
Enter a positive integer: 12345

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 1
Even_dig = 2
Odd_dig = 3
aaryanrajput@aaryans-macbook ~ % ./day13.c
Enter a positive integer: 12345

MENU
1. Count even and odd digits
2. Find sum of even and odd digits
Enter your choice: 2
Sum_even = 6
Sum_odd = 9
aaryanrajput@aaryans-macbook ~ %
```

Compilation and Execution successful

Additional Testing

For `24680` with Choice `1`, the program produced **Even_dig = 5, Odd_dig = 0**. For `13579` with Choice `2`, it produced **Sum_even = 0, Sum_odd = 25**. Both matched the expected results.

Final Result: All 4 prepared test cases passed successfully.